

Exact Markov Elimination for Relational Current Tracking

Anonymous

Abstract—Glory first be to The Holy One, Blessed be He, from whom all good things come. We study position-only tracking with a finite Gaussian current prior and an observation-driven acquisition rate. Position is the exact integral of orthonormal cell currents; a five-component correlation-length mixture retains uncertainty about their continuation. We derive a four-state realization per spatial coordinate and mixture component that reproduces the original finite midpoint covariance, noisy anchor, observation likelihood, and posterior. This eliminates dense current-history covariance from streaming updates without changing the represented trajectory model. On an M4 CPU, a 128-cell observation update decreases from 452.7 to 12.2 microseconds; the corresponding NumPy realization already achieves 23.4 microseconds. Across 480 matched acquisition cases, the implementations produce identical acquisition times and boundary-trigger decisions, with position and forecast differences below 9×10^{-12} meters. A separate retained synthetic comparison includes constant-velocity and constant-acceleration Kalman filters, interacting multiple-model filters, a robust variant, and a three-dimensional turn unscented Kalman filter. At low noise with 65 observations, the current model’s two-second endpoint RMSE is 2.958 meters, versus 3.269 for robust IMM; the advantage does not persist decisively in every noise and sampling condition. Adaptive density reduces acquisition count and improves tracking after a settling interval. We also give explicit model-conditional uncertainty growth and a sharp bounded-acceleration error certificate common to all compared estimators; no superior worst-case breakdown rate is established. Exact elimination changes the computational balance enough that scheduling, rather than assimilation, becomes the dominant adaptive-loop cost.

Index Terms—Target tracking, Gaussian processes, Kalman filtering, Matérn covariance, state-space realization, adaptive sampling, exact elimination.

I. INTRODUCTION

A position tracker uses accumulated observations to correct its current belief and to continue that belief beyond the last measurement. These two operations depend on the same temporal law. A representation that exposes all past and future current coefficients makes that dependence explicit, but it can conceal a much smaller sufficient filtering state. The computational question is therefore precise: can the calculation be reorganized while preserving the prior, evidence, and resulting decisions?

We consider a finite current construction on a declared time interval. Each Cartesian position is an anchor plus the exact integral of a piecewise-constant current. The current covariance combines a shared constant component

with a Matérn-3/2 component, and a fixed mixture of five correlation lengths represents uncertainty about continuation scale. The first implementation conditioned a dense covariance in either real current coordinates or a unitary Zak representation. Both coordinates produce the same posterior when the full covariance is retained.

The present work derives and implements a four-state realization of this *finite* prior. Two states realize the sampled Matérn current; the other two carry the shared constant current and accumulated position. The resulting filter retains every length hypothesis and its exact Gaussian evidence. The reformulation changes the order in which latent history is eliminated. It does not replace the finite current by a continuously integrated process, discard mixture components, or introduce a new correction-selection rule.

There are three distinct empirical questions. First, how does the declared current prior compare with existing tracking implementations? Second, what accuracy and acquisition-count tradeoff follows when a boundary estimate increases observation density? Third, what computational cost can be removed from exactly the same estimation calculation? We answer these questions with separate protocols and retained records. We do not combine their timings or error statistics into a single ranking.

The contributions are a finite-model equivalence theorem including a noisy anchor and a severed-current ablation; an implementation that shares spatial covariance, evidence normalization, and cached cell propagation; a matched evaluation of numerical equivalence and acquisition cost; and explicit uncertainty-growth bounds that separate model-conditional posterior statements from physical prediction-error certificates. The empirical comparison supports competitiveness on the tested synthetic families, with uncertainty intervals and failure-to-dominate conditions reported alongside the favorable case. No real-flight or universal tracking-optimality claim is made.

II. RELATION TO EXISTING TRACKING METHODS

Kalman filtering supplies recursive inference for linear Gaussian state models [1]. A temporal Gaussian process can also admit a finite Markov realization. Hartikainen and Särkkä give exact state-space constructions for temporal Matérn covariance functions and discuss nonuniform observations [2]. Parallel temporal GP algorithms further exploit associative filtering and smoothing structure [3]. These are established connections. The four-state realization below specializes them to a fixed midpoint-current

model and its cellwise integral; we do not claim the general GP-state-space equivalence as new.

An interacting multiple-model filter (IMM) instead mixes dynamically switching motion hypotheses [4]. Its model set and transition law determine which maneuvers can be represented. Unscented filtering propagates nonlinear state uncertainty through sigma points [5]. Our measured turn UKF estimates a continuous three-dimensional turn vector, whereas our IMM uses a finite catalog of axis-aligned turn generators in its state coordinates. This makes the UKF a useful control for the IMM catalog's limitations.

Current tracking research includes comparisons of fixed and adaptive multiple-model transition laws [6], IMM filtering and smoothing on manifolds [7], and feedback particle filters on matrix Lie groups [8]. Their observation contracts and algorithms differ from the local comparators evaluated here. In particular, our robust IMM is a specified contamination-likelihood implementation, not a reproduction of every adaptive robust IMM method. The recent studies provide context, not additional measured baselines.

The length mixture in this paper has a *fixed latent length per trajectory hypothesis*. Its weights update by marginal likelihood; its states are not mixed through an IMM transition matrix. It consequently represents uncertainty about temporal correlation rather than a Markov sequence of switching maneuver labels. “Relational” refers here to the specified covariance between cell currents. No general nonlinear relational denoiser is implied by that name.

III. FINITE CURRENT AND OBSERVATION MODEL

Let $[0, T]$ be a declared representation interval, N the number of equal cells, and $h = T/N$. The endpoint T is known to the representation; it is not an observation of future motion. All queries lie in this interval.

Definition 1 (Finite current position). *For one spatial coordinate, let a be an anchor error and v_i the current value in cell i , $0 \leq i < N$. Relative to the first measured position y_0 , define*

$$x(t) = y_0 + a + \sum_{i=0}^{N-1} v_i \text{clip}(t - ih, 0, h). \quad (1)$$

The orthonormal cell basis and its coefficients are

$$\phi_i(t) = h^{-1/2} \mathbf{1}_{[ih, (i+1)h)}(t), \quad c_i = \sqrt{h} v_i.$$

The current is therefore $\sum_i c_i \phi_i$.

For each correlation length ℓ , let $\lambda_\ell = \sqrt{3}/\ell$ and prescribe the stationary cell-current covariance

$$\begin{aligned} \text{Cov}(v_i, v_j | \ell) &= q_g + q_w(1 + \lambda_\ell d_{ij})e^{-\lambda_\ell d_{ij}}, \\ d_{ij} &= |i - j|h. \end{aligned} \quad (2)$$

All experiments fix $q_g = q_w = 4 \text{ m}^2 \text{ s}^{-2}$ and

$$\ell \in \mathcal{L} = \{0.3, 0.7, 1.5, 3, 6\} \text{ s}, \quad \Pr(\ell) = 1/5. \quad (3)$$

The covariance is evaluated at cell midpoints. Their common half-cell offset cancels in d_{ij} . The coefficient covariance is h times (2); increasing N does not silently rescale current units.

Observations are complete three-dimensional positions

$$\mathbf{y}_j = \mathbf{x}(t_j) + \boldsymbol{\epsilon}_j, \quad \boldsymbol{\epsilon}_j \sim \mathcal{N}(0, \sigma^2 I_3), \quad (4)$$

with known $\sigma > 0$ and independent errors. The history begins at $t_0 = 0$ and has strictly increasing arrival times. Conditioning a flat absolute-position prior on $\mathbf{y}_0$ gives $\mathbf{a} \sim \mathcal{N}(0, \sigma^2 I_3)$ relative to that observation. This anchor remains uncertain. The later sensor errors are independent of it; $\mathbf{y}_0$ is not assimilated a second time. Conditional on ℓ , the three axes have independent, identical temporal priors.

A dense coefficient state has $N + 1$ entries per axis. Its measurement row is

$$B(t) = \left[1, \left\{ \frac{\text{clip}(t - ih, 0, h)}{\sqrt{h}} \right\}_{i=0}^{N-1} \right]. \quad (5)$$

An invertible coordinate change preserves the observation law when the state, covariance, and measurement rows all transform consistently. In particular, a unitary Zak matrix U gives covariance UPU^* and row BU^* . The complex representation must retain the real-signal conjugacy constraints; it does not justify a proper-complex observation likelihood.

IV. EXACT FOUR-STATE REALIZATION

Write $v_i = g + w_i$, with independent $g \sim \mathcal{N}(0, q_g)$ and a Matérn-3/2 current sequence w_i . Introduce the normalized derivative $r = w'/\lambda_\ell$ of an auxiliary continuous process, sampled at the current midpoints. This auxiliary process realizes the covariance; the position remains the finite integral in (1).

Lemma 1 (Sampled current realization). *Put $\alpha = \lambda_\ell h$ and*

$$A_\ell = e^{-\alpha} \begin{bmatrix} 1 + \alpha & \alpha \\ -\alpha & 1 - \alpha \end{bmatrix}, \quad Q_\ell = q_w(I_2 - A_\ell A_\ell^\top). \quad (6)$$

For $z_i = (w_i, r_i)^\top$, the stationary recursion

$$z_{i+1} = A_\ell z_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, Q_\ell), \quad z_0 \sim \mathcal{N}(0, q_w I_2) \quad (7)$$

has the Matérn part of (2) as its first-coordinate covariance. The innovations are independent across cells.

Proof. The continuous generator $D = \lambda_\ell \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ satisfies

$$D(q_w I_2) + (q_w I_2)D^\top = -\text{diag}(0, 4\lambda_\ell q_w). \quad (8)$$

Thus Q_ℓ is the integral of $e^{Du} \text{diag}(0, 4\lambda_\ell q_w) e^{D^\top u}$ over $0 \leq u \leq h$ and is positive semidefinite. Exponentiating Dh gives (6). The nilpotent part of $D + \lambda_\ell I_2$ squares to zero, so

$$A_\ell^n = e^{-n\alpha} \begin{bmatrix} 1 + n\alpha & n\alpha \\ -n\alpha & 1 - n\alpha \end{bmatrix}. \quad (9)$$

Stationarity follows from $A_\ell(q_w I_2)A_\ell^\top + Q_\ell = q_w I_2$. The (1, 1) entry of $q_w A_\ell^n$ is $q_w(1 + n\alpha)e^{-n\alpha}$, as required. $\square$

Let p_i be the accumulated position relative to y_0 at the start of cell i . Define $s_i = (p_i, g, w_i, r_i)^\top$. One cell advances by

$$s_{i+1} = F_\ell s_i + \xi_i, \quad (10)$$

$$F_\ell = \begin{bmatrix} 1 & h & h & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & (A_\ell)_{11} & (A_\ell)_{12} \\ 0 & 0 & (A_\ell)_{21} & (A_\ell)_{22} \end{bmatrix},$$

$$\text{Cov}(\xi_i) = \text{diag}(0, 0, Q_\ell),$$

$$\text{Cov}(s_0) = \text{diag}(\sigma^2, q_g, q_w, q_w). \quad (11)$$

Here the block notation in (11) places Q_ℓ in the last two coordinates. At $t \in [ih, (i+1)h]$, let $\tau = t - ih$. The observation row and position are

$$H(t) = [1, \tau, \tau, 0], \quad x(t) = y_0 + H(t)s_i. \quad (12)$$

At $t = T$, use $i = N - 1$ and $\tau = h$. Multiple observations within a cell use the same latent state with different rows; no artificial current innovation is introduced between them.

Theorem 1 (Finite-model equivalence). *For fixed N, T, σ and $\mathcal{L}$, the dense model (1)–(2) and the realization (10)–(12) define identical joint position laws conditional on the initial observation y_0 . After any finite increasing observation history satisfying (4), they therefore have identical component posterior laws, marginal likelihoods, mixture weights, and position posterior laws at all represented query times.*

Proof. Lemma 1, together with the independent g , gives exactly (2). The initialization gives the same independent anchor error. The first row of F_ℓ yields $p_i = a + h \sum_{j < i} (g + w_j)$, so (12) is exactly (1). These are equal linear functions of equal Gaussian prior laws. Adding the same independent observation errors preserves the joint law of observations and queries. Gaussian conditioning gives the same conditional distributions and evidence for each ℓ . Bayes' rule with the same length prior then gives the same mixture. $\square$

Corollary 1 (Filtering sufficiency). *Given the component posterior over s_i and the length weights after the latest observation, all current and future position marginals can be computed without retaining the dense coefficient posterior.*

Proof. Future current innovations in (7) are independent of the past conditional on z_i . The state also contains the accumulated position and shared constant current. Conditioning on these variables separates future propagation from eliminated history. $\square$

This theorem concerns the finite model at a fixed N . It is not a claim that changing N leaves the prior unchanged, nor that a continuous Matérn integral has the same covariance. The auxiliary derivative in Lemma 1 supplies the Markov realization while the finite cell integral supplies the actual position law.

A. Severed Current Covariance

For an ablation time t_s , the dense reference zeros covariance between currents whose midpoints lie on opposite sides of t_s . Let $k_s = \#\{i : (i+1/2)h < t_s\}$. After integrating cell $k_s - 1$, reset (g, w, r) to an independent Gaussian with covariance $\text{diag}(q_g, q_w, q_w)$, preserving accumulated position. The reset replaces cross-group current covariance by zero and reproduces the dense ablation exactly. For $k_s = 0$ or N , there is no internal reset. This operation removes a continuation connection deliberately; it is not an alternative estimate selected from future truth.

V. SHARED CONDITIONING AND COMPILED EVALUATION

Let $M_\ell \in \mathbb{R}^{4 \times 3}$ contain the component state means and $P_\ell \in \mathbb{R}^{4 \times 4}$ its common axis covariance immediately before an observation. Compute

$$b_\ell = P_\ell H^\top, \quad S_\ell = H b_\ell + \sigma^2, \quad (13)$$

$$e_\ell = y - y_0 - H M_\ell, \quad (13)$$

$$M_\ell^+ = M_\ell + (b_\ell / S_\ell) e_\ell, \quad P_\ell^+ = P_\ell - b_\ell b_\ell^\top / S_\ell. \quad (14)$$

The innovation e_ℓ is a row vector. The same rank-one covariance update serves all three axes. The log-evidence increment is

$$\Delta L_\ell = -\frac{1}{2} \left(3 \log(2\pi S_\ell) + \|e_\ell\|^2 / S_\ell \right). \quad (15)$$

Initialize $L_\ell = -\log |\mathcal{L}|$ and normalize once after each observation by $\pi_\ell = e^{L_\ell} / \sum_k e^{L_k}$, using log-sum-exp. Readouts reuse the weights and total log evidence.

For component query means $\mu_\ell(t)$ and scalar variances $V_\ell(t)$, the reported moments are

$$\bar{\mu}(t) = \sum_\ell \pi_\ell \mu_\ell(t), \quad (16)$$

$$\Sigma(t) = \sum_\ell \pi_\ell [V_\ell(t) I_3 + \delta_\ell(t) \delta_\ell(t)^\top], \quad \delta_\ell = \mu_\ell - \bar{\mu}. \quad (17)$$

The between-component term is retained. The result need not be spatially isotropic even though each component's within-axis covariance is shared.

Proposition 1 (Likelihood refinement and coordinates). *Assimilating one Gaussian likelihood in m equal fractional powers gives the same posterior as assimilating it once. An invertible state-coordinate change with consistent covariance and measurement transforms gives the same observation evidence and physical position posterior.*

Proof. For a Gaussian likelihood $\mathcal{L}$, the product of m factors $\mathcal{L}^{1/m}$ equals $\mathcal{L}$. Each normalized Gaussian factor has noise variance $m\sigma^2$; its normalization differs from that of $\mathcal{L}^{1/m}$ only by a state-independent constant. Posterior normalization removes that constant. Model evidence must still be computed under the original likelihood, as in (15), rather than by treating the factors as additional physical observations. A coordinate change leaves the physical observation random variable unchanged, proving the second statement. $\square$

Integer-cell propagation uses shared precomputed tables

$$\begin{aligned} F_{\ell,n} &= F_{\ell}^n, & Q_{\ell,0} &= 0, \\ Q_{\ell,n+1} &= F_{\ell} Q_{\ell,n} F_{\ell}^{\top} + \text{Cov}(\xi_i). \end{aligned} \quad (18)$$

A jump of n cells applies $F_{\ell,n}$ and $Q_{\ell,n}$ once. Each query propagates a copy of the filtering state, so evaluation does not change subsequent estimates or acquisition decisions. The compiled kernel fuses branch propagation, rank-one updates, and evidence normalization into a single update call; a separate call evaluates all requested current/future marginal moments. Double precision is used without fast-math.

With $K = |\mathcal{L}|$ fixed, streaming update work is $O(K)$ after the $O(KN)$ table construction. At $N = 128$, covariance entries per branch decrease from $129^2 = 16641$ to $4^2 = 16$. The five filtering means and covariances occupy 1120 bytes in double precision, excluding weights, object overhead, and shared tables. This is a statement about the filtering state, not total program memory.

The implementation retains the observation history for compatibility. Queries earlier than the last observation, and exports of the full coefficient posterior, lazily replay the dense reference. Their cache is invalidated by a new observation. These operations retain dense costs. The transition cache is bounded and shared between trackers, and the retained input history grows with the observation count. The fast API returns position marginals; it does not return a certified continuous boundary-crossing probability from those marginals alone.

VI. CAUSAL ACQUISITION DENSITY

The acquisition experiment changes sampling cadence while retaining one posterior and one update law. At each acquired position, the tracker is updated once. The first acquisition time for which the updated estimated first coordinate is nonnegative becomes a latched trigger $t_{\star}$. No true crossing time enters this trigger. For $t \geq t_{\star}$, the requested rate is

$$r(t) = r_0 + (r_1 - r_0) \left(1 - e^{-(t-t_{\star})/\tau_r}\right), \quad (19)$$

with $r_0 = 2$ Hz, $\tau_r = 1$ s, and $r_1 \in \{4, 8, 16, 32\}$ Hz. Before the trigger $r(t) = r_0$. Two controls use constant 2 and 32 Hz rates.

Irregularity is introduced in phase rather than by rounding times to a common grid. Draw $u_j \sim \text{Uniform}(0.8, 1.2)$ independently and choose the next time from

$$\int_{t_j}^{t_{j+1}} r(s) ds = u_j. \quad (20)$$

The retained implementation uses 44 bisection steps for the ramped integral. This scheduler is unchanged in the optimized replay. The rate command is continuous, but individual observations and posterior updates remain discrete. The covariance-weighted innovation in (14) supplies the correction; no separate witness gate or boundary-dependent filter selection is applied.

Proposition 2 (Causal policy equivalence). *For the same initial observation and random phase/noise streams, exact posterior equivalence implies identical acquisition sequences under (19)–(20), provided the same numerical scheduling map is used in the exact-arithmetic comparison.*

Proof. The policies initially use the same observation and schedule. If the acquired history agrees through step j , Theorem 1 gives the same corrected position. The trigger condition and rate therefore agree. The same next phase gives the same next time, and the same observation law and noise draw give the same next observation. Induction proves the claim. $\square$

Floating-point reordering can change a sign decision exactly at a threshold. Consequently exact decision agreement is measured as well as proved algebraically; no universal bitwise claim follows from the theorem.

VII. EXPERIMENTAL PROTOCOLS

A. Matched Forecast Comparison

The retained screen contains five motion families: straight motion, tilted helices, spatial figure eights, smooth stop/restart, and motion approaching changing waypoints. Truth uses analytic positions or a separate fine-step Cartesian simulation. Random orientations and translations avoid a preferred global observation frame. Future waypoint commands are unavailable to all estimators.

Six fresh seeds per family give 30 trajectories in each of three paired conditions: C1 uses 65 observations with $\sigma = 0.35$ m; C2 uses 65 with $\sigma = 1.4$ m; C3 uses 9 with $\sigma = 0.35$ m. Histories span $[0, 6]$ s; queries are 20 times from 6.1 through 8 s. The 65 observations are evenly spaced; the sparse condition takes every eighth history observation. Noise draws are paired across conditions. The current representation uses $T = 8$ s and $N = 64$. Eight methods give 720 method/case runs. Future truth is used only by the evaluator.

The five conventional controls are local literature-family implementations. CV and CA use linear kinematics; IMM mixes CV, CA, and six fixed turns of magnitude 0.65 rad/s about signed Cartesian axes. Its mixing time scale is 3 s. Robust IMM uses the same model set with a two-component observation-noise mixture having probabilities (0.97, 0.03) and variances ($\sigma^2, 100\sigma^2$), recombining the noise branches by moments. The continuous-turn UKF uses $(\alpha, \beta, \kappa) = (1, 2, 0)$ and estimates a full 3-D turn vector. All conventional filters include the pre-existing augmented position/bias state. Their priors are therefore not identical to the current prior.

Configurations were frozen from a separate ten-case development stage, with three candidate configurations per method and objective tracking MSE plus 0.25 times two-second forecast MSE. Selected process intensities are $q = 1$ for CV and robust IMM, and $q = 0.1$ for CA and IMM; the UKF turn/speed diffusivities are 0.75/0.30. Only supplied sensor σ changes in this screen. The five current lengths and current variances were declared before inspecting this

screen and were not selected on its future truth. Unequal development histories and differing model classes limit claims of globally optimal parameter matching.

The current model produces analytic forecast moments. Conventional forecasts retain the existing 2048-path Monte Carlo implementation, with separate fixed forecast seeds. Thus their estimated means include finite Monte Carlo error. These historical forecast workloads are not used as matched-kernel timing comparisons.

The primary error is the Euclidean 3-D position error at 8 s. Reported RMSE is the square root of mean squared endpoint distance over the 30 cases. Paired percentile intervals use 4000 resamples of the six seeds within each of the five fixed families. They are pointwise intervals, not simultaneous tests or uncertainty over unseen motion families.

B. Acquisition Accuracy and Matched Replay

Boundary-crossing fixtures use the same five families with eight new seeds and two noise levels. Their first coordinate is replaced by $2(t - 6) + 0.3 \sin(0.7t + \varphi)$, with seed-dependent phase; the remaining coordinates follow the existing generated family. Truth is represented on a fixed 0.01 s grid over $[0, 14]$ s and linearly interpolated for queries. These are explicitly modified crossing fixtures, not the unchanged trajectories of Section VII-A.

The tracker uses $T = 14$ s, $N = 128$. Six acquisition policies give 480 policy/case runs. Current position and two-second forecasts are evaluated at the same 0.1 s grid over $[0, 12]$ s for every policy. The true crossing is used only to define evaluation windows: the first second, seconds 3–6 after crossing, and the full first six post-crossing seconds, intersected with the evaluation interval. Per-case MSEs are averaged with equal case weight before taking their square root.

The exact-optimization study replays all 480 cases through the archived dense implementation and compiled realization, giving 960 matched runs. Run order alternates by family/seed to reduce systematic order effects. Sample arrays, decisions, current estimates, forecasts, and uncertainty radius diagnostics are compared. There is no timing-based or accuracy-based tracker selection during acquisition.

C. Cost Controls

Microbenchmarks use an identical 160-observation irregular stream, 31 repeats after one warm-up, double precision, and one BLAS thread on the M4 Mini CPU. Dense 512-cell configurations use five repeats. The four implementations are dense Zak, dense real current, four-state NumPy, and four-state compiled. Each repeat times updates, current-position readouts, and two-query forecasts separately. Cold initialization clears the transition cache; warm initialization is measured in the acquisition replay. No CPU instruction counters were collected: the evidence is algebraic state/work reduction and measured elapsed time.

A separate small CV cost control has two states per axis, a shared isotropic covariance, white-acceleration intensity $q = 1$, and no mixture or bias state. It is implemented in NumPy and compiled C++, with the same irregular observation stream and Python interface style. This control establishes a simple implementation cost scale. It is not the augmented CV model in the earlier accuracy screen and carries no matched-accuracy claim.

Total online acquisition cost includes initialization, assimilation, readouts, and scheduling. Diagnostic-only pre-update probes and offline truth/error scoring are excluded. Sensor energy, communications, bandwidth, and acquisition latency are not assigned a monetary or CPU cost. All timing claims refer to these retained implementations and this host rather than a real-time deadline guarantee.

VIII. UNCERTAINTY GROWTH AND CONDITIONAL ERROR BOUNDS

Exact inference does not by itself bound error against an arbitrary trajectory. We distinguish a posterior statement conditional on the current prior, a physical error certificate conditional on motion and sensor bounds, and empirical degradation under changed conditions.

A. Growth Under the Declared Current Prior

Fix a realized observation history ending at t_b and a represented query $t_b + u \leq T$. In component ℓ , write the posterior position variance at t_b as $V_{b,\ell}$ per coordinate. Let $a_i(u)$ be the length of overlap between cell i and $[t_b, t_b + u]$, so $0 \leq a_i \leq h$ and $\sum_i a_i = u$. Define

$$W_\ell(u) = q_g u^2 + q_w \sum_{i,j} a_i a_j \rho_{|i-j|}, \quad (21)$$

$$\rho_k = (1 + k\alpha)e^{-k\alpha}, \quad \alpha = \lambda_\ell h. \quad (22)$$

Gaussian conditioning cannot increase the covariance of this fixed linear increment. Cauchy–Schwarz on its posterior covariance with the current position therefore gives

$$V_\ell(t_b + u) \leq (\sqrt{V_{b,\ell}} + \sqrt{W_\ell(u)})^2. \quad (23)$$

The nonnegative Toeplitz kernel has row sum at most

$$C_\alpha = 1 + 2 \left(\frac{z}{1-z} + \frac{\alpha z}{(1-z)^2} \right), \quad z = e^{-\alpha}. \quad (24)$$

Using $\rho_k \leq 1$ and the row-sum operator bound with $\sum_i a_i^2 \leq hu$ proves

$$W_\ell(u) \leq q_g u^2 + q_w \min\{u^2, hC_\alpha u\}. \quad (25)$$

Thus component standard deviation has an explicit at-most-linear envelope in forecast horizon, within the declared interval. This is a property of the chosen stochastic prior, not a bound on arbitrary motion.

Let $\mu_\ell(u)$ and $V_\ell(u)I_3$ be the component position moments, and $\bar{\mu}(u)$ the mixture mean. For $0 < \delta < 1$, the ball of radius

$$R_\delta(u) = \max_\ell \left\{ \|\mu_\ell(u) - \bar{\mu}(u)\| + \sqrt{6V_\ell(u) \log(6/\delta)} \right\} \quad (26)$$

about $\bar{\mu}(u)$ has posterior probability at least $1-\delta$ under the declared mixture model. Indeed, the coordinate Gaussian tail bound and a union bound give the stated component radius; the triangle inequality places every component ball inside this ball. Averaging the conditional probabilities over the length weights preserves $1-\delta$. This includes between-component mean separation, unlike using component variance alone. It is a single-query posterior statement, not a simultaneous continuous-time tube or a guarantee for the synthetic truth families. Exact elimination preserves this bound and its inputs.

B. A Sharp Bound on Actual Prediction Error

A separate, estimator-independent assumption supplies a physical certificate. Suppose $\mathbf{x}'$ is absolutely continuous and $\|\mathbf{x}''(t)\| \leq A$ almost everywhere over the observation and forecast interval. Suppose two observations at $t_a < t_b$ satisfy $\|\mathbf{y}_i - \mathbf{x}(t_i)\| \leq \varepsilon_i$. These assumptions do not follow from a nominal sensor standard deviation or from the current prior.

Theorem 2 (Two-observation continuation bound). *Put $\Delta = t_b - t_a > 0$ and $u \geq 0$. Define*

$$\mathbf{c}(u) = \mathbf{y}_b + \frac{u}{\Delta}(\mathbf{y}_b - \mathbf{y}_a), \quad (27)$$

$$r(u) = \left(1 + \frac{u}{\Delta}\right)\varepsilon_b + \frac{u}{\Delta}\varepsilon_a + \frac{A}{2}u(\Delta + u). \quad (28)$$

Then $\|\mathbf{x}(t_b + u) - \mathbf{c}(u)\| \leq r(u)$. The constants are sharp even in one dimension. For any estimator output $\mathbf{m}_j(u)$, including current, CV, CA, IMM, robust IMM, or UKF, a valid bound is

$$\|\mathbf{x}(t_b + u) - \mathbf{m}_j(u)\| \leq r(u) + \|\mathbf{m}_j(u) - \mathbf{c}(u)\|. \quad (29)$$

Proof. Translate t_b to zero. The error of extrapolating the two true positions is the integral of acceleration against the nonnegative kernel

$$K(s) = \begin{cases} u(s + \Delta)/\Delta, & -\Delta \leq s \leq 0, \\ u - s, & 0 \leq s \leq u. \end{cases}$$

Its integral is $u(\Delta + u)/2$. The two measurement coefficients have absolute values $1 + u/\Delta$ and u/Δ , giving (28) by the triangle inequality. In one dimension, $x(t) = At^2/2$, $y_a = x(-\Delta) + \varepsilon_a$, and $y_b = x(0) - \varepsilon_b$ attain equality. A final triangle inequality gives (29). $\square$

For equal error limits ε , the growth and noise sensitivity are

$$r(u) = \varepsilon + bu + \frac{1}{2}Au^2, \quad b = 2\varepsilon/\Delta + A\Delta/2, \quad (30)$$

$$r'(u) = b + Au, \quad \partial_\varepsilon r = 1 + 2u/\Delta. \quad (31)$$

For a tolerated radius $d > \varepsilon$ and $A > 0$, this pair certificate expires at the explicit horizon

$$u_d = \frac{2(d - \varepsilon)}{b + \sqrt{b^2 + 2A(d - \varepsilon)}}. \quad (32)$$

This is loss of the stated certificate, not a prediction that error actually reaches d . For $A = 0$ and $b > 0$,

$u_d = (d - \varepsilon)/b$. For a different estimator center, the offset in (29) must also be included before asserting a deadline. The noise-amplification and curvature terms balance at $\Delta = 2\sqrt{\varepsilon/A}$ when both parameters are positive. A shorter observation gap alone can therefore worsen this two-point bound.

Every observation pair supplies a ball; their intersection retains all such constraints. Appending observations while retaining the old balls cannot enlarge that set. Sample count alone supplies no universal $n^{-1/2}$ error law: a shared unknown sensor bias need not average away. For a fixed history of n isotropic Gaussian observations with known marginal standard deviations, choosing

$$\varepsilon_i = \sigma_i \sqrt{6 \log(6n/\delta)} \quad (33)$$

makes all observation limits hold with probability at least $1-\delta$, by coordinate tails and a union bound. Independence is unnecessary for this union bound, but the marginal error laws must be valid. On that event, the physical balls hold simultaneously at all future times for which the acceleration assumption holds. Adaptive stopping requires an appropriate conditional or anytime observation guarantee; the fixed- n argument alone does not justify it.

Even perfect knowledge of past position and velocity cannot remove the worst-case radius $Au^2/2$: two continuations with opposite constant accelerations share the entire past and separate by Au^2 . Any common prediction errs by at least half that separation on one continuation. Without a future-motion restriction there is no finite uniform bound. Consequently none of the compared estimators has a better worst-case horizon order under this common assumption. Differences in their recentered bounds remain possible, but have not been measured here.

The finite current model itself has velocity jumps at cell boundaries, and the acquisition fixtures use linear interpolation on a fine grid. Neither literally satisfies the bounded-acceleration premise in general. We therefore do not attach an acceleration constant or certified physical radius to those experiments. The physical theorem states what additional motion and sensor information would be needed. Conversely, a smaller model covariance, in this method or in a Kalman-family comparator, does not establish a stronger physical guarantee.

IX. RESULTS

A. Comparison with the Earlier Tracking Controls

Table I includes every method in the matched current screen. At low noise with 65 observations, relational-current endpoint RMSE is 2.958 m, compared with 3.750 for CV, 6.319 for CA, 3.443 for IMM, 3.269 for robust IMM, and 4.243 for the turn UKF. The paired current/robust IMM ratio is 0.905 with interval $[0.837, 0.981]$; the current/IMM ratio is 0.859 with interval $[0.779, 0.952]$.

The same ordering does not hold in every condition. At high noise, robust IMM reaches 4.806 m versus 4.891 m for the current mixture; their ratio is 1.018 with interval $[0.921, 1.124]$. In the sparse low-noise condition, current

TABLE I
TWO-SECOND ENDPOINT RMSE (m) ON THE MATCHED FORECAST SCREEN. C1: 65 OBSERVATIONS, $\sigma = 0.35$ m; C2: 65 OBSERVATIONS, $\sigma = 1.4$ m; C3: 9 OBSERVATIONS, $\sigma = 0.35$ m. EACH COLUMN CONTAINS 30 PAIRED CASES.

Method	C1	C2	C3
Relational current	2.958	4.891	3.788
Severed current	4.868	5.240	4.932
Single length, 1.5 s	3.556	4.762	3.886
CV Kalman	3.750	5.357	4.355
CA Kalman	6.319	7.811	6.140
IMM	3.443	5.014	4.263
Robust IMM	3.269	4.806	3.940
3-D turn UKF	4.243	5.229	4.653

TABLE II
ENDPOINTS CONTAINED IN NOMINAL 95% MOMENT ELLIPSOIDS. THESE ARE EMPIRICAL DIAGNOSTICS, NOT EXACT MIXTURE COVERAGE GUARANTEES.

Method	C1	C2	C3
Relational current	30/30	28/30	30/30
Severed current	30/30	30/30	30/30
Single length, 1.5 s	30/30	30/30	30/30
CV Kalman	28/30	26/30	27/30
CA Kalman	17/30	16/30	16/30
IMM	17/30	19/30	17/30
Robust IMM	30/30	30/30	30/30
3-D turn UKF	21/30	15/30	17/30

RMSE is 3.788 m versus 3.940 m for robust IMM, with ratio 0.961 and interval [0.906, 1.021]. These intervals do not establish a consistent advantage over robust IMM. Figure 1 shows all five comparator ratios; the numerical intervals appear in Appendix A.

The severed-current ablation increases C1 error to 4.868 m. A single 1.5 s length gives 3.556 m. The corresponding paired mixture/ablation ratios are 0.608 [0.537, 0.675] and 0.832 [0.757, 0.908]. Severing establishes that the declared past-future covariance matters for these forecasts. It does not establish uniqueness of that continuation law. At high noise the single-length model reaches 4.762 m, slightly below the mixture's 4.891 m.

Table II reports containment using $(x - \bar{\mu})^T \Sigma^{-1} (x - \bar{\mu}) \leq 7.814727903$. The current mixture contains 30/30, 28/30, and 30/30 endpoints, with mean equal-volume ellipsoid radii 5.77, 8.42, and 6.61 m. A Gaussian mixture is not itself generally Gaussian, and these trajectories were not drawn from the current prior. Consequently the chi-square threshold does not certify 95% posterior coverage here. Good endpoint RMSE and large uncertainty regions must be assessed together.

B. Measured Degradation Under Noise and Sparsity

Table III compares each method with its own C1 error. The current mixture's RMSE rises 65.4% when sensor standard deviation quadruples, and 28.1% when observations decrease from 65 to 9. Robust IMM rises 47.0% and 20.5%, respectively. The current model thus has larger relative degradation point estimates despite its lower C1

TABLE III
WITHIN-METHOD ENDPOINT RMSE INFLATION RELATIVE TO C1, WITH POINTWISE 95% PAIRED PERCENTILE INTERVALS (4000 RESAMPLES WITHIN EACH OF FIVE FIXED FAMILIES). CONDITIONS ARE DEFINED IN TABLE I. AN INFLATION OF 1.654 MEANS 65.4% MORE RMSE.

Method	C2/C1 [interval]	C3/C1 [interval]
Relational current	1.654 [1.473, 1.857]	1.281 [1.202, 1.369]
CV Kalman	1.428 [1.301, 1.569]	1.161 [1.118, 1.211]
CA Kalman	1.236 [1.089, 1.419]	0.972 [0.915, 1.036]
IMM	1.456 [1.306, 1.598]	1.238 [1.094, 1.415]
Robust IMM	1.470 [1.324, 1.615]	1.205 [1.146, 1.283]
3-D turn UKF	1.232 [1.159, 1.312]	1.097 [1.045, 1.153]

TABLE IV
ACQUISITION COUNT AND POST-CROSSING ACCURACY. COUNTS COVER 12 s; RMSE USES THE FIRST SIX POST-CROSSING SECONDS. RATES ARE IN HZ, ERRORS IN METERS.

σ	Rate profile	Samples	Position	2 s forecast
0.35	Constant 2	24.5	0.681	3.249
0.35	2 to 4	34.0	0.544	3.042
0.35	2 to 8	53.0	0.441	2.804
0.35	2 to 16	91.3	0.363	2.636
0.35	2 to 32	167.8	0.307	2.493
0.35	Constant 32	383.9	0.209	2.331
1.4	Constant 2	24.5	1.833	4.395
1.4	2 to 4	34.0	1.537	4.147
1.4	2 to 8	52.9	1.291	3.867
1.4	2 to 16	91.0	1.084	3.686
1.4	2 to 32	167.2	0.918	3.464
1.4	Constant 32	383.9	0.646	3.209

error and its still slightly lower C3 error. A lower starting error can also increase a relative-growth ratio. These are two finite changes, not fitted scaling exponents or formal breakdown laws; overlapping pointwise intervals do not test the difference between methods' growth. The earlier geometric prototype's degradation results concern a different estimator and cannot establish a current-model rate.

C. Acquisition Benefit and Settling

Added observations reduce current error and two-second prediction error on these crossing fixtures (Table IV and Fig. 2). At $\sigma = 0.35$ m, adaptive 2-to-32 Hz uses 167.8 samples on average, compared with 383.9 for constant 32 Hz. Whole post-crossing position RMSE is 0.307 versus 0.209 m, so the reduced acquisition count has a measurable accuracy cost over that complete window.

After three seconds, adaptive and always-dense tracking are close (Table V). Their later RMSE ratio is 0.996 with interval [0.958, 1.034] at low noise and 1.001 with interval [0.951, 1.053] at high noise. The first-second ratios are 2.706 and 2.359, respectively. Dense acquisition has already accumulated more information when the boundary is crossed; adaptive acquisition pays for its sparse approach with this settling interval.

Individual mixture corrections also become smaller as history grows. At low noise, the later mean correction decreases from 0.702 m under constant 2 Hz to 0.077 m under adaptive 2-to-32 Hz. The measured projection of

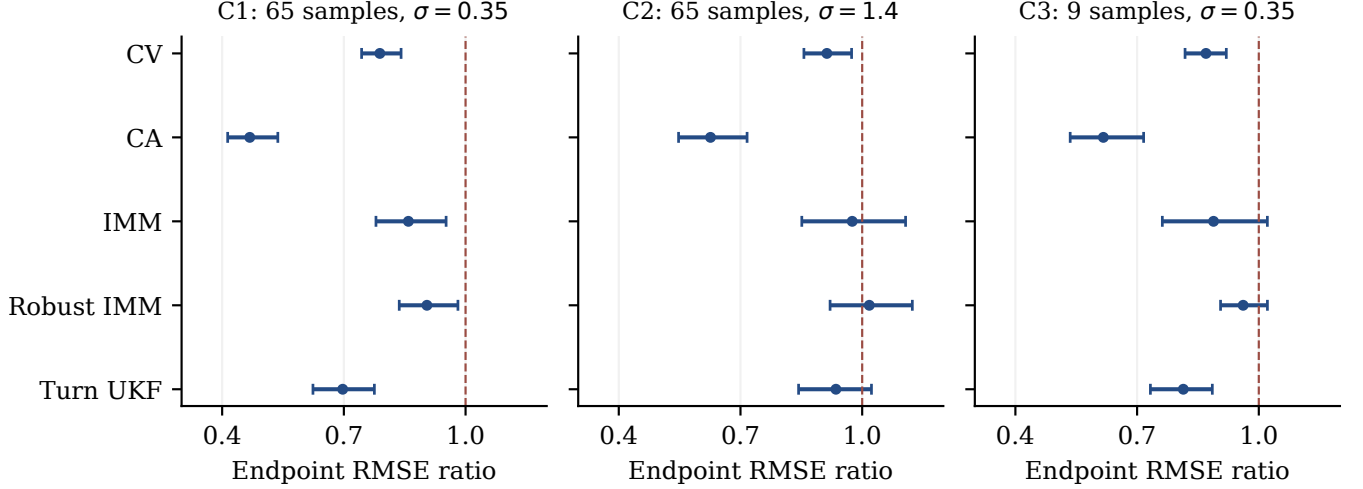


Fig. 1. Current-model endpoint RMSE divided by comparator endpoint RMSE. Values below one favor the current model. Bars are pointwise 95% paired percentile intervals from 4000 within-family resamples. The same 30 cases support each condition; the intervals do not cover arbitrary motion families or simultaneous comparisons.

TABLE V

SETTLING AFTER THE TRUE CROSSING: POSITION RMSE (M). THE TRUE CROSSING DEFINES EVALUATION WINDOWS ONLY.

σ	Rate profile	First second	Seconds 3–6
0.35	Constant 2	0.729	0.687
0.35	2 to 32	0.563	0.209
0.35	Constant 32	0.208	0.210
1.4	Constant 2	1.858	1.853
1.4	2 to 32	1.550	0.645
1.4	Constant 32	0.657	0.644

TABLE VI

128-CELL MICROBENCHMARK: MEDIAN MICROSECONDS PER CALL. FORECAST REQUESTS RETURN CURRENT AND TWO-SECOND-AHEAD MARGINAL MOMENTS.

Implementation	Update	Position	Forecast
Dense Zak	452.68	43.50	136.80
Dense real current	250.62	30.99	79.87
Four-state NumPy	23.38	2.37	22.97
Four-state compiled	12.18	2.32	22.58

the correction onto the innovation decreases from 0.716 to 0.129. This projection describes the mixture correction; it does not impose a clipping coefficient.

D. Exactness and Cost

All 480 dense/compiled acquisition pairs have identical trigger times, counts, timestamps, and observation arrays. Maximum absolute differences are 7.29×10^{-12} m for evaluated current positions, 8.13×10^{-12} m for two-second forecasts, 6.26×10^{-12} m for acquisition-event estimates, and 1.37×10^{-13} m for reported radius diagnostics. These are numerical equivalence measurements, not improvements in tracking accuracy or universal bitwise identities.

The compiled update is 37.2 times faster than the original at 128 cells (Table VI). Moving the dense computation

TABLE VII

WHOLE ONLINE COMPUTE OVER A 12 S TRAJECTORY, $\sigma = 0.35$ M. MEANS OVER 40 CASES; COSTS INCLUDE SCHEDULING AND READOUT, BUT NOT SENSOR ACQUISITION. EACH PAIR FOLLOWS IDENTICAL ACQUISITION DECISIONS.

Rate profile	Dense (ms)	Compiled (ms)	Speedup
Constant 2	33.87	3.54	9.57
2 to 32	114.47	12.55	9.12
Constant 32	218.18	9.58	22.78

to real current coordinates yields only 1.8 times, whereas the four-state NumPy realization already yields 19.4 times. The state elimination is therefore the main reduction. At 32, 128, and 512 cells, compiled updates take 11.99, 12.18, and 12.45 microseconds, versus 150.63, 452.68, and 5129.44 for dense Zak. The 128-cell cold initialization is 0.357 ms for the compiled realization and 4.744 ms for dense Zak; with shared tables, warm acquisition initialization is approximately 0.028 ms.

The small compiled CV cost control takes 5.39 microseconds per update and 11.16 per two-query forecast. The five-component current model costs 2.26 and 2.02 times those operations, respectively. The NumPy CV control takes 7.28 and 10.39 microseconds. Compilation therefore does not remove all interface overhead, and neither comparison proves a lower bound on achievable native latency. The CV control is independently checked against its NumPy equations over 160 irregular updates and 320 queries.

Whole-loop effects differ from isolated updates (Table VII and Fig. 3). Adaptive 2-to-32 Hz falls from 114.47 to 12.55 ms, a 9.12-fold reduction. Assimilation falls from 78.09 to 2.29 ms, while the unchanged scheduler takes 6.73 ms and becomes the largest component. Constant 32 Hz falls from 218.18 to 9.58 ms. Thus the optimized adaptive loop costs *more CPU* than always-dense acquisition in

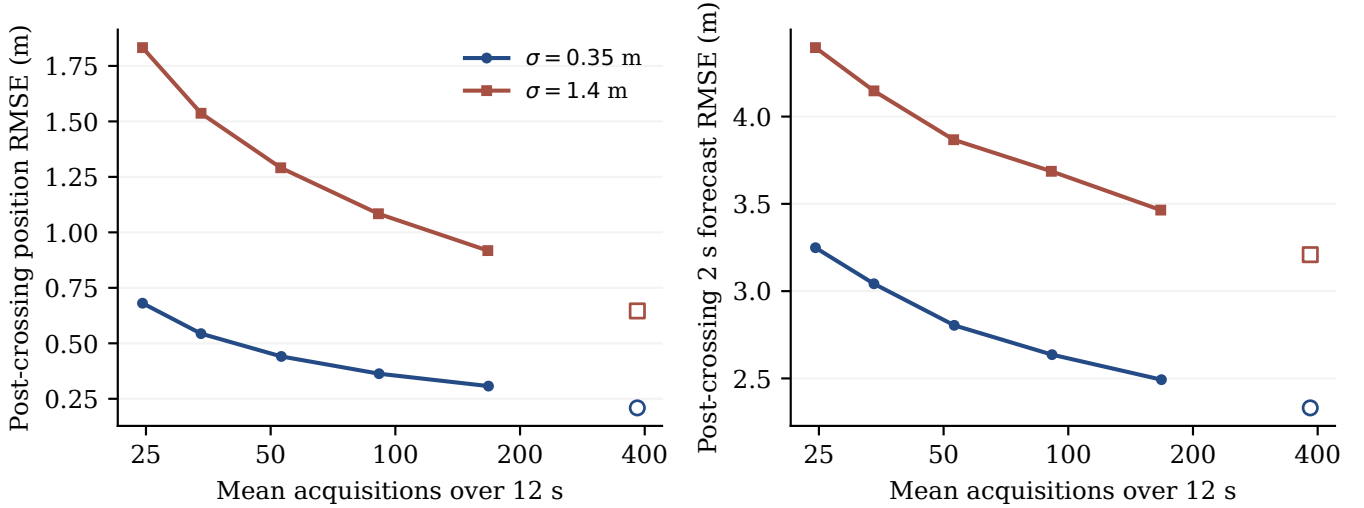


Fig. 2. Acquisition count versus post-crossing accuracy, averaged over 40 cases per noise level. Connected filled markers run from constant 2 Hz through the four adaptive target rates. Open markers are constant 32 Hz. All policies use the same physical evaluation grid. The same accuracy values are retained by exact elimination; the figure shows sampling effects rather than an optimization-induced accuracy change.

this implementation, despite using only 43.7% as many samples. This reversal is a consequence of inexpensive filtering and nontrivial rate inversion. It must not be hidden by reporting assimilation cost alone. The high-noise runs show the same cost pattern.

X. INTERPRETATION AND LIMITATIONS

The computational and statistical claims have different scopes. Theorem 1 establishes equality for the declared finite Gaussian model. The synthetic comparisons ask whether that model is useful outside its own prior distribution. Favorable low-noise forecast errors cannot extend the theorem into a universal prediction guarantee. Unknown future commands, correlated sensor drift, and noise misspecification remain substantive limitations.

The measured current model is competitive with the selected conventional implementations, but it does not decisively dominate robust IMM in every condition. The earlier baseline forecasts also carry Monte Carlo error. A more extensive benchmark would require external trajectories, additional contemporary implementations, matched sensor assumptions, and a common analytical or adequately converged forecast interface. The separate simple CV timing control cannot supply those missing accuracy comparisons.

The representation horizon is fixed, not periodically wrapped or extended implicitly. Five 64-to-128-cell checks in the forecast screen change endpoint means by less than 0.022 m; four 128-to-256-cell acquisition checks change position RMSE by less than 0.00013 m and forecast RMSE by less than 0.00084 m. These limited checks do not prove a uniform continuum limit. Very fine cells also require care when forming Q_ℓ by subtracting nearly equal matrices.

Twenty-three focused tests pass: 17 existing relational/acquisition checks and six realization checks covering both NumPy and native backends. They include ker-

nel reconstruction, covariance stationarity, 8/32/128-cell streams, two coordinate systems, six server configurations, within-cell and irregular observations, evidence, fractional updates, historical queries, full coefficient exports, and cache invalidation. The archived dense oracle has the same source hash as the original acquisition implementation. Five additional physical-certificate tests pass, including exact-rational sharpness, a finite-turn path, retained constraints, recentering, and invalid-input rejection. These test the conditional certificate separately. Tests and replay establish the stated implementation fidelity; they are not a formal floating-point proof.

XI. CONCLUSION

A dense finite current history can be eliminated exactly when its covariance admits a small Markov realization. For the specified constant-plus-Matérn-3/2 current model, accumulated position, constant current, local current, and normalized current derivative suffice. Shared spatial conditioning and evidence preserve the five-component posterior while reducing a 128-cell update from 452.7 to 12.2 microseconds on the measured host.

The resulting computational scale is within a small factor of a simple compiled CV filter, while the retained forecast comparison shows a low-noise benefit over the measured robust IMM and no decisive benefit in every other condition. Adaptive acquisition provides a separate count-accuracy tradeoff with a settling interval. Once exact elimination makes assimilation inexpensive, scheduling can determine the CPU cost of that tradeoff. The representation, prediction law, numerical fidelity, and acquisition policy must therefore be evaluated separately and then accounted for together.

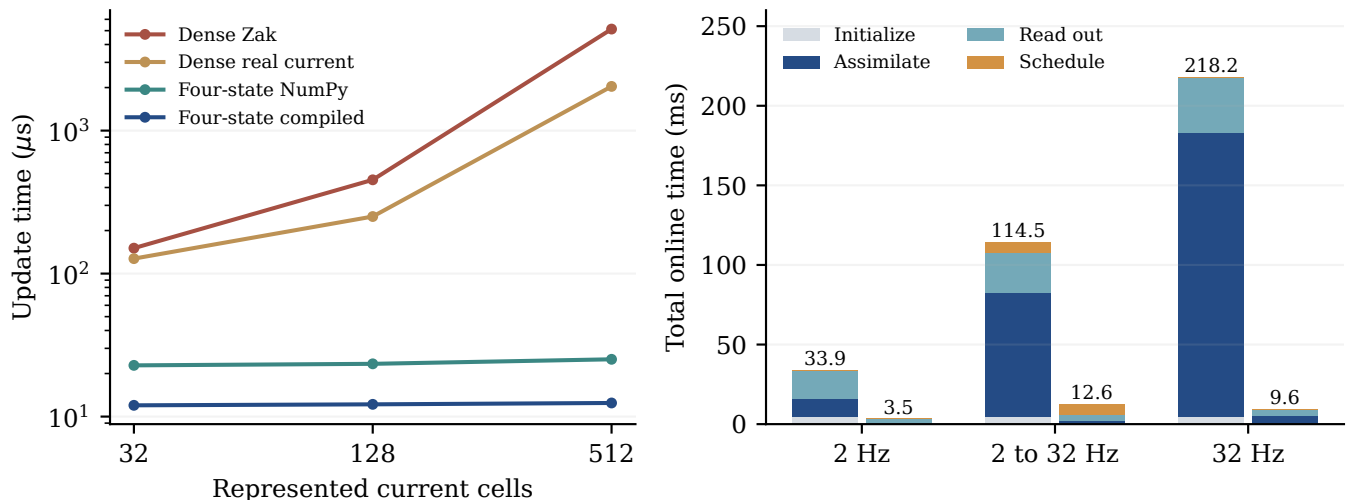


Fig. 3. Left: update scaling with represented cell count, in double precision. Right: whole online acquisition cost at $\sigma = 0.35$ m; each pair shows dense (left) and compiled (right). Scheduling is unchanged and dominates the optimized adaptive loop. Microbenchmarks use 31 repeats except 512-cell dense cases (five); whole-loop bars average 40 matched cases per policy.

APPENDIX A COMPLETE PAIRED FORECAST INTERVALS

Table VIII gives the paired current/comparator RMSE ratios under the protocol of Section VII-A. IMM and robust IMM intervals retain the original reported bootstrap draws. CV, CA, and UKF intervals are computed from the same retained case errors with the same 4000-resample within-family procedure and a separately fixed resampling seed. No method is retuned. All intervals are pointwise.

TABLE VIII
CURRENT/COMPARATOR ENDPOINT RMSE RATIOS AND 95% PAIRED PERCENTILE INTERVALS. CONDITION DEFINITIONS ARE IN TABLE I.

Condition	Comparator	Ratio	Interval
C1	CV Kalman	0.789	[0.744, 0.841]
C1	CA Kalman	0.468	[0.414, 0.537]
C1	IMM	0.859	[0.779, 0.952]
C1	Robust IMM	0.905	[0.837, 0.981]
C1	3-D turn UKF	0.697	[0.624, 0.775]
C2	CV Kalman	0.913	[0.856, 0.974]
C2	CA Kalman	0.626	[0.547, 0.716]
C2	IMM	0.975	[0.851, 1.107]
C2	Robust IMM	1.018	[0.921, 1.124]
C2	3-D turn UKF	0.935	[0.843, 1.023]
C3	CV Kalman	0.870	[0.818, 0.920]
C3	CA Kalman	0.617	[0.535, 0.716]
C3	IMM	0.888	[0.762, 1.021]
C3	Robust IMM	0.961	[0.906, 1.021]
C3	3-D turn UKF	0.814	[0.733, 0.885]

APPENDIX B EARLIER GEOMETRIC-TRANSPORT CONFIRMATION

For completeness, Table IX records the earlier continuous-turn geometric-transport confirmation alongside the same conventional filter families. This is a *different cohort and method*: 80 trajectories, eight seeds in each of five family/two corruption strata, 6144 filtering

particles for geometric transport, and 256 forecast paths. It uses irregular observation intervals, approximately two-second rolling forecasts, and corruption episodes including drift, outliers, noise-scale changes, and missing samples. The current model of Section III was not evaluated in this cohort, so its errors cannot be compared directly with this table.

The geometric prototype transports a particle distribution over speed, direction, turn, position, and bias through finite rotation maps. Its 6144-particle confirmation obtained a 3.6% endpoint RMSE reduction against robust IMM, with a paired interval of $[-2.1\%, 8.5\%]$. That result did not establish an advantage over the strongest robust control. Its crossing Brier score was also worse. These findings motivate keeping accuracy, uncertainty, and boundary decisions distinct; they are not additional evidence of current-model superiority.

TABLE IX
SEPARATE GEOMETRIC CONFIRMATION COHORT.
TRACKING/FORECAST RMSE AND ENERGY SCORE ARE IN METERS.
BRIER SCORE CONCERNS SAMPLED-PATH CROSSING OF $x = 0$. THE
CURRENT MODEL IS ABSENT FROM THIS COHORT.

Method	Track	~ 2 s	Energy	Brier
CV Kalman	0.664	4.495	2.825	0.088
CA Kalman	0.698	6.488	4.280	0.107
IMM	0.648	4.122	2.410	0.085
Robust IMM	0.610	3.624	2.250	0.084
3-D turn UKF	0.683	3.757	2.525	0.097
Geometric transport	0.684	3.495	2.222	0.099

APPENDIX C REPRODUCTION AND ARTIFACT MAP

The accompanying evidence archive retains the forecast, acquisition, optimization, and geometric-confirmation records separately, with a SHA-256 source manifest. The

manuscript’s tables and vector figures are generated from these records by `generate_assets.py`; rendering performs no fitting or new estimator selection. The TeX bundle includes all generated table inputs and figures, and uses standard packages without shell escape or machine-specific fonts. It compiles without access to the experiment repository.

The repository experiment directory is `experiments/civilian_transport`. Its `relational_reference.py` preserves the dense oracle; `relational_markov.py` specifies the four-state algebra; `relational_native.py` and `native/markov.cpp` expose and implement the compiled operations. The runner `run_optimization_m4.sh` builds the optional kernels, runs the 23 focused tests, performs the 480-case paired replay and microbenchmark, and copies the result files back from the compute host. The standalone cost control is `cost_controls.py`.

The earlier forecast screen is reproduced by `run_relational_m4.sh`; acquisition accuracy by `run_acquisition_m4.sh`; and the separate geometric confirmation by `run_confirmation_m4.sh`. Re-running through the promoted streaming class changes computation cost but not the specified finite prior. Original timing evidence remains archived rather than being silently replaced by a rerun.

TABLE X

128-CELL UPDATE VARIABILITY: MEDIAN AND 10TH–90TH PERCENTILES OF REPEAT-AVERAGE MICROSECONDS PER OBSERVATION.

Implementation	Median	10th–90th percentile
Dense Zak	452.68	[448.11, 455.26]
Dense real current	250.62	[247.49, 253.04]
Four-state NumPy	23.38	[23.26, 24.01]
Four-state compiled	12.18	[11.99, 12.29]

REFERENCES

- [1] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, 1960, doi: 10.1115/1.3662552.
- [2] J. Hartikainen and S. Särkkä, “Kalman filtering and smoothing solutions to temporal Gaussian process regression models,” in *Proc. IEEE Int. Workshop Mach. Learn. Signal Process.*, 2010, pp. 379–384, doi: 10.1109/MLSP.2010.5589113.
- [3] A. Corenflos, Z. Zhao, and S. Särkkä, “Temporal Gaussian process regression in logarithmic time,” arXiv:2102.09964, 2021. [Online]. Available: <https://arxiv.org/abs/2102.09964>
- [4] H. A. P. Blom and Y. Bar-Shalom, “The interacting multiple model algorithm for systems with Markovian switching coefficients,” *IEEE Trans. Autom. Control*, vol. 33, no. 8, pp. 780–783, 1988, doi: 10.1109/9.1299.
- [5] S. J. Julier and J. K. Uhlmann, “Unscented filtering and nonlinear estimation,” *Proc. IEEE*, vol. 92, no. 3, pp. 401–422, 2004, doi: 10.1109/JPROC.2003.823141.
- [6] F. F. Lizzio, E. I. Trombetta, E. Capello, and Y. Fujisaki, “Comparative evaluation of multiple-model Kalman filters for highly maneuvering UAV tracking,” *Appl. Sci.*, vol. 16, no. 5, Art. no. 2377, 2026, doi: 10.3390/app16052377.
- [7] T. L. Koller and U. Frese, “The interacting multiple model filter and smoother on boxplus-manifolds,” *Sensors*, vol. 21, no. 12, Art. no. 4164, 2021, doi: 10.3390/s21124164.
- [8] C. Zhang, A. Taghvaei, and P. G. Mehta, “Feedback particle filter on matrix Lie groups,” arXiv:1701.02416, 2017. [Online]. Available: <https://arxiv.org/abs/1701.02416>